

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I G.Amrutha(192511250) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

amrutha

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- li. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- lii. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\neg$ Wumpus [x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Case Study 1 — Smart Medical Diagnosis System

(a) Representation using Propositional Logic

Let the following propositional symbols represent the facts:

Symbol	Meaning
F	Patient A has Fever

Symbol	Meaning
C	Patient A has Cough
R	Patient A has Rash
B	Patient A has Breathlessness
Flu	Patient A has Possible Flu
Measles	Patient A has Possible Measles
Pneumonia	Patient A has Possible Pneumonia

Rules

Rule I:

If Fever AND Cough $\rightarrow$ Possible Flu

Rule II:

Rule III:

Facts

Therefore, the complete knowledge base is:

(b) Forward Chaining

Forward chaining starts with known facts and repeatedly applies rules whose conditions are satisfied.

Initial Facts

Step 1 — Apply Rule I

Rule I:

We know:

and

Therefore:

Derived fact: Patient A may have Flu.

Step 2 — Apply Rule II

Rule II:

We have:

but there is **no fact that** .

Therefore:

No Measles conclusion can be derived.

Step 3 — Apply Rule III

Rule III:

We know:

and

Therefore:

Derived fact: Patient A may have Pneumonia.

Final Forward-Chaining Conclusions

Measles cannot be concluded.

Inference Sequence

↓

Rule I

↓

↓

Rule III

↓

(c) Backward Chaining for Pneumonia

Goal

Search for a rule that concludes Pneumonia.

Rule III is:

Therefore, the goal is reduced to two subgoals:

↓

Now check the knowledge base.

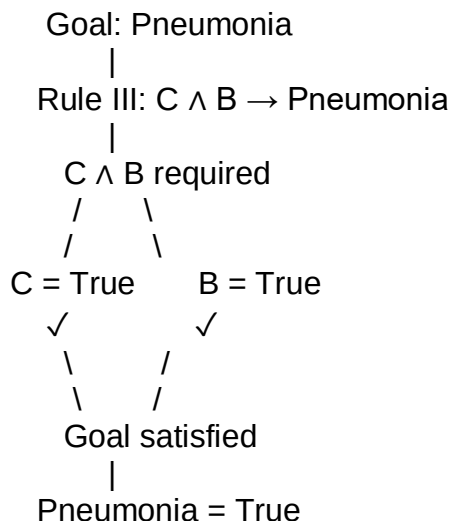
We have:

and

Both subgoals are satisfied.

Therefore:

Goal Reduction Tree



Conclusion

Backward chaining confirms:

Case Study 2 — Autonomous Traffic Management Agent

Given FOL Knowledge Base

Rule 1

Rule 2

Rule 3

Facts

(a) Unification with Rule 1

Rule 1:

We have:

and

We need:

Therefore the substitution is:

First Unification

matches:

giving:

Second Unification

becomes:

which matches the fact.

Therefore:

Applying the substitution to Rule 1:

Since both premises are true:

(b) Resolution Proof of Proceed(CarA)

We want to prove:

Using resolution refutation, assume the opposite:
Then derive a contradiction.

Step 1 — Convert Rule 1 to CNF

Original:

Using:

we get:

Call this **C1**.

Step 2 — Rule 2

Rule 2:

This produces two clauses:

C2

and

C3.

Step 3 — Rule 3

Original:

CNF:

Call this C4.

Step 4 — Relevant Facts

C5:

C6:

C7:

C8:

Negated goal:

C9:

Resolution Steps**Resolution 1**

C1 with C5:

and

Substitution:

Therefore:

Resolution 2

Resolve with C6:

Therefore:

Resolution 3

Use C2:

with:

Therefore:

Resolution 4

C4:

Use:

Substitution:

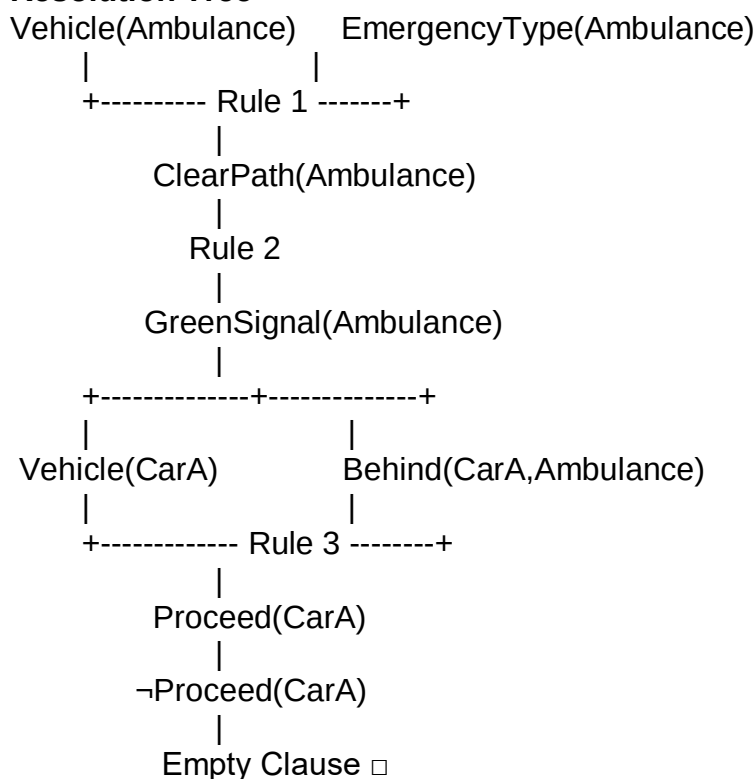
Therefore:

Resolution 5 — Contradiction

We also have the negated goal:

Resolving:
 with:
 gives:
 where is the empty clause.
 Therefore:

Resolution Tree



(c) Two Simultaneous Emergency Vehicles

The current knowledge base can handle **more than one emergency vehicle individually**, because Rule 1 is universally quantified:

So if we have:

and:

Rule 1 can derive:

and

Then Rule 2 derives:

and

However, there is a limitation.

The KB does not contain rules for **conflict management** when both emergency vehicles require the same road or signal.

Additional rules could include:

or:

We also need facts such as:

and possibly:

Conclusion

The current KB can reason about multiple vehicles, but it is **not sufficient for simultaneous emergency conflicts**. Priority, collision avoidance, signal conflict, and route allocation rules are needed.

Case Study 3 — Agricultural Expert Reasoning System

Given KB

P1

P2

P3

Facts

(a) Conversion to CNF

P1

Original:

Eliminate implication

Using:

we get:

This is already CNF.

P2

Original:

Eliminate implication:

Move negation inward using De Morgan's law:

Therefore:

This is CNF.

P3

Original:

Eliminate implication:

Apply De Morgan's law:

Therefore:

Facts in CNF

Fact 1:

Fact 2:

Complete CNF KB

(b) Resolution Refutation for ApplyDripMethod

Goal

Negate the goal:

Add it to the KB.

Resolution 1

C1:

Fact:

Resolve:

Resolution 2

C2:

We have:

Resolve:

Resolution 3

We have:

Resolve with:

Therefore:

Resolution 4 — Contradiction

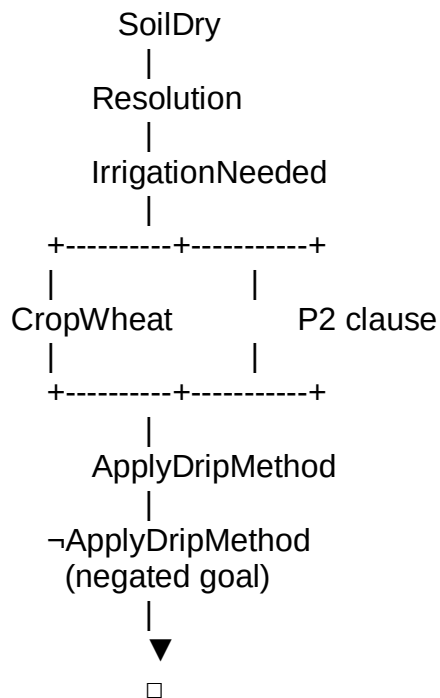
Negated goal:

and derived:

Resolve:

Therefore:

Resolution Proof Tree



Therefore, the theorem is proved.

(c) Remove SoilDry

Suppose:

is removed from the KB.

Remaining facts:

We still have:

but we no longer know SoilDry is true.

Therefore, we **cannot derive**:

Consequently, P2 cannot fire:

Therefore:

Does CropAtRisk follow?

P3 is:

Although ApplyDripMethod cannot be proved, that does **not** mean:
is true.

This is the important distinction between:

- **Not proved**
- **Proved false**

The absence of ApplyDripMethod does not allow us to infer ¬ApplyDripMethod.

Therefore:

Final Analysis

Situation

SoilDry present

IrrigationNeeded + CropWheat ApplyDripMethod derived

SoilDry removed

IrrigationNeeded cannot be derived

ApplyDripMethod

Cannot be derived

¬ApplyDripMethod

Not established

CropAtRisk

Cannot be derived